Weighted interval scheduling
COMS20017 (Algorithms and Data)

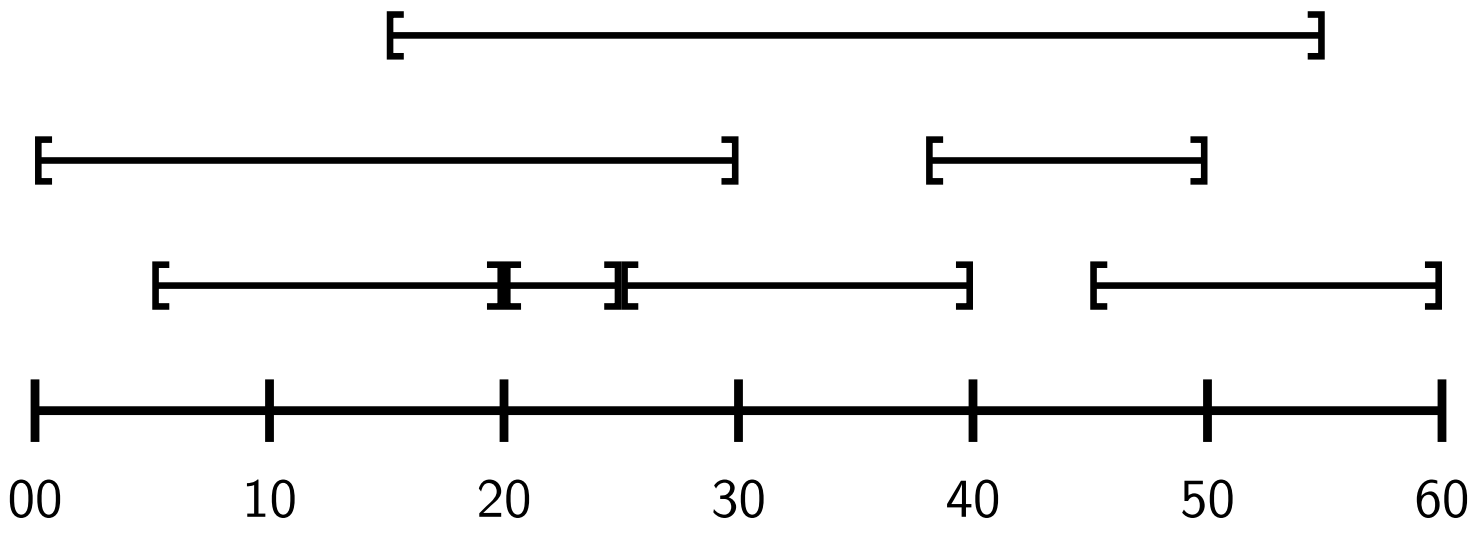
John Lapinskas, University of Bristol

Unweighted interval scheduling (recap from week 2)

Motivation: A satellite imaging service wants to use its camera to fill as many orders as possible, but it can only take one picture at once.

Input: A set of intervals, e.g.
$\mathcal{R} = \{(0,30),(5,20),(15,55),(20,25),(25,40),(38,50),(45,60)\}$.

Output: A maximum **compatible** set $\mathcal{R}' \subseteq \mathcal{R}$ — that is, for all $(s,f),(s',f') \in \mathcal{R}'$, we have either $s',f' \geq f$ or $s',f' \leq s$.



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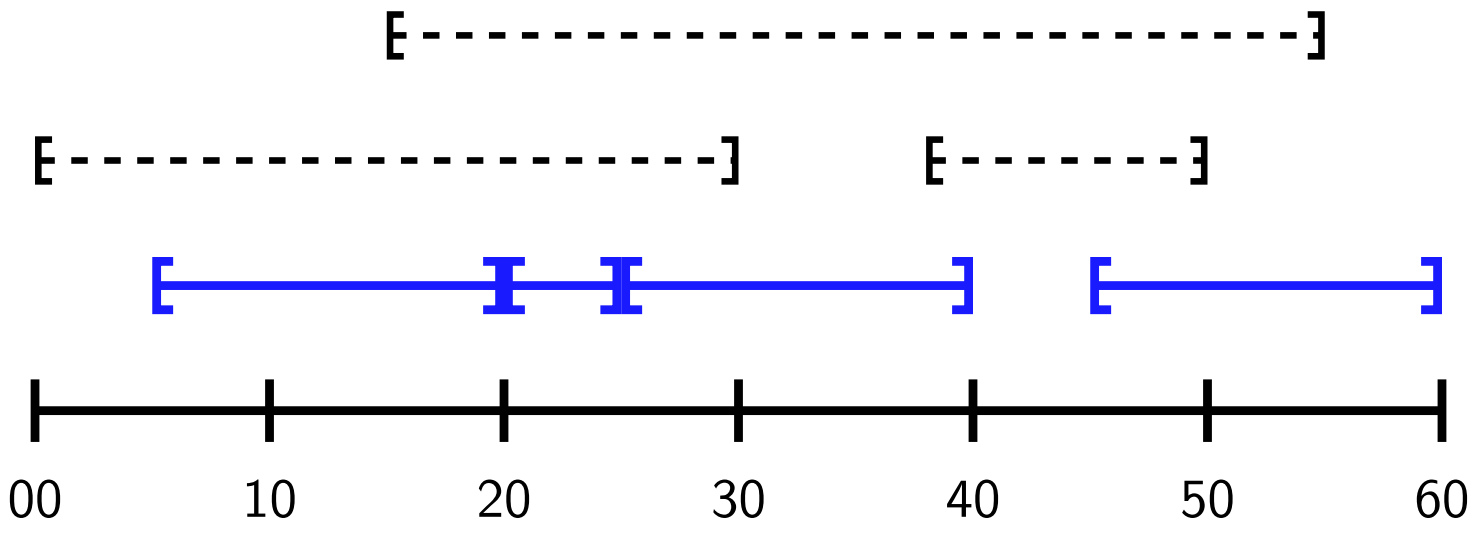

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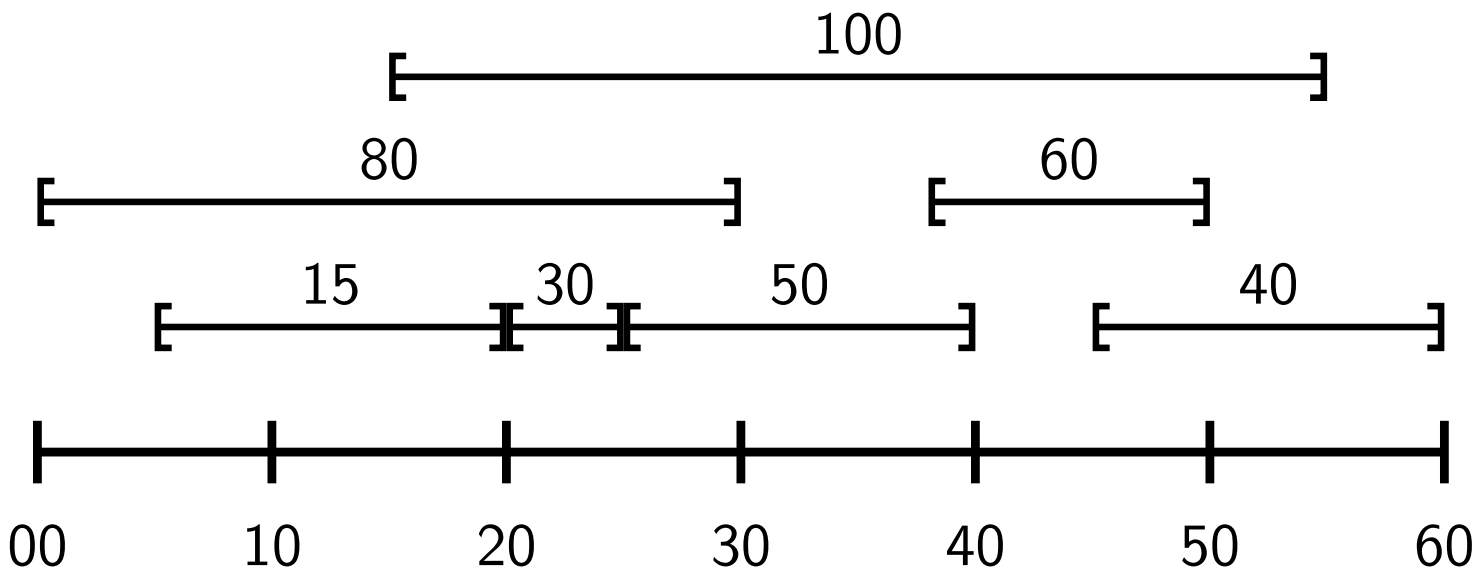

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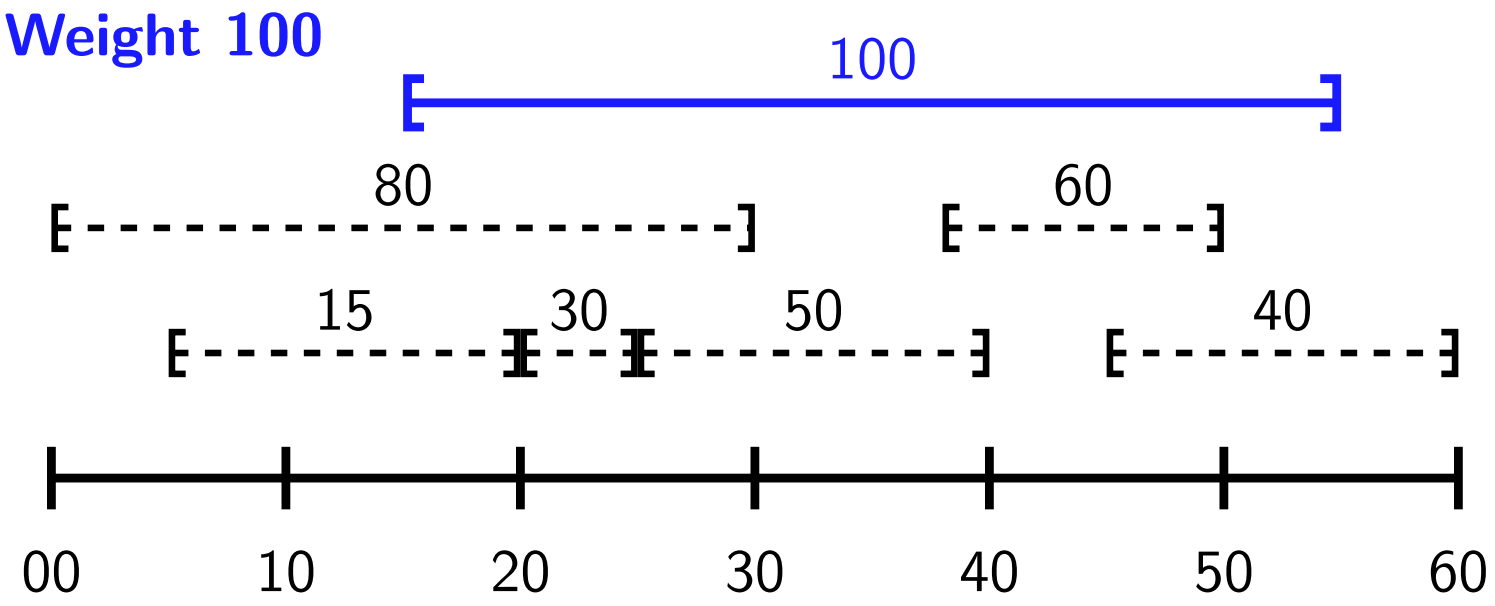

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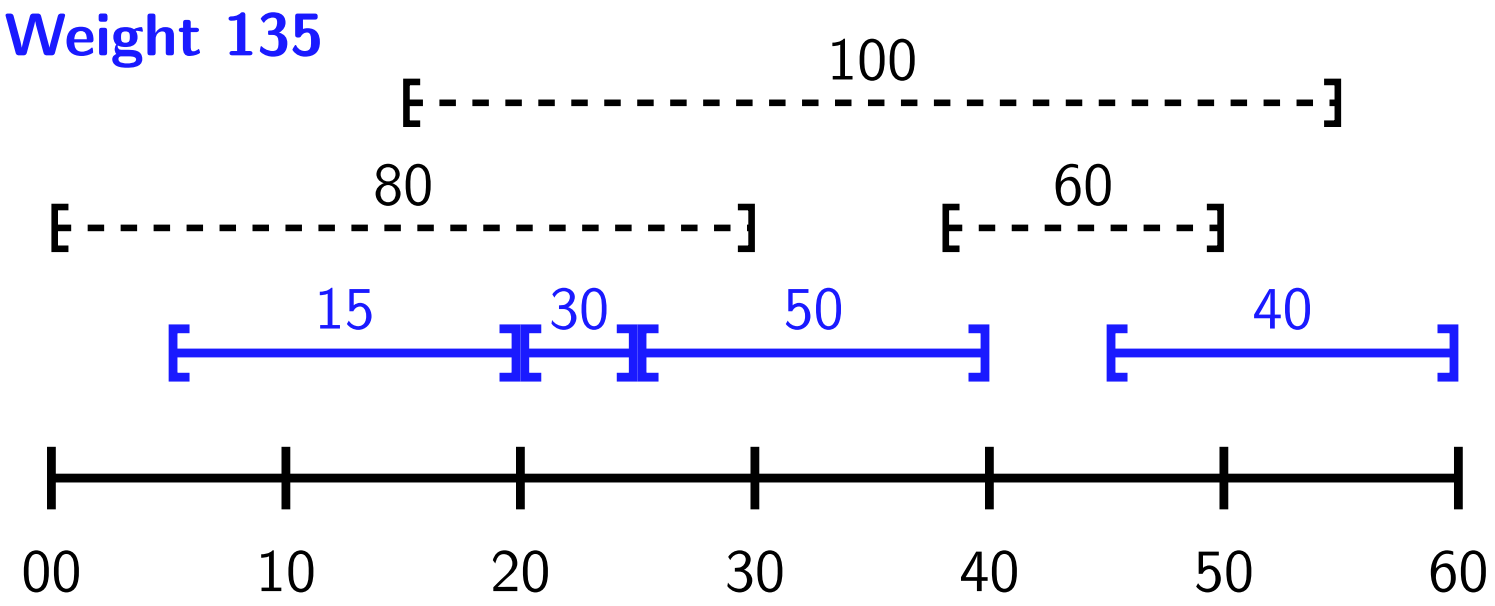

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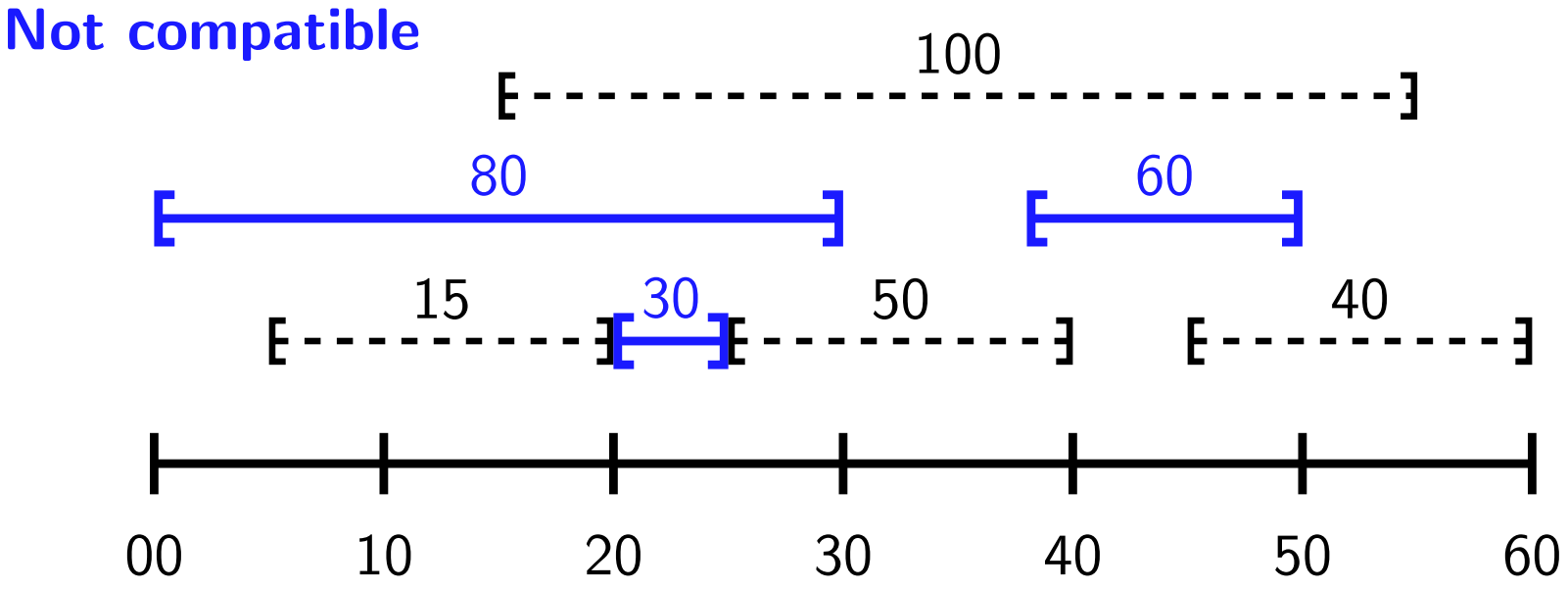

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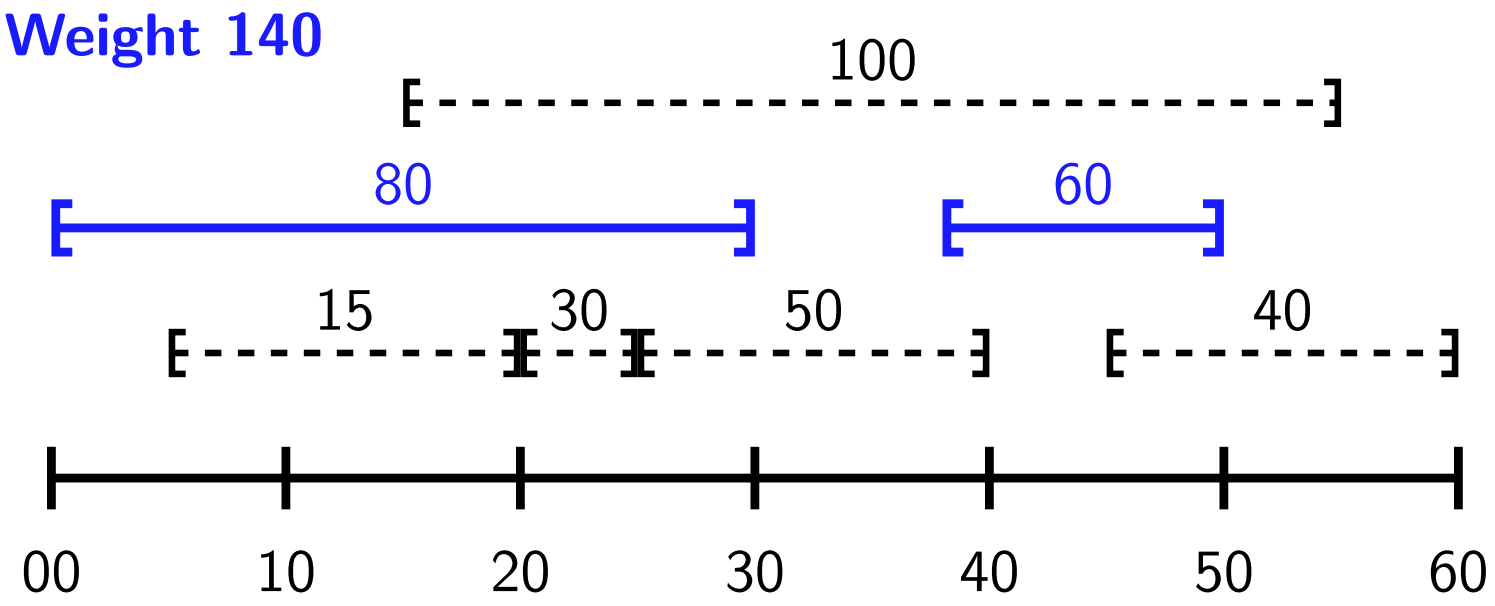

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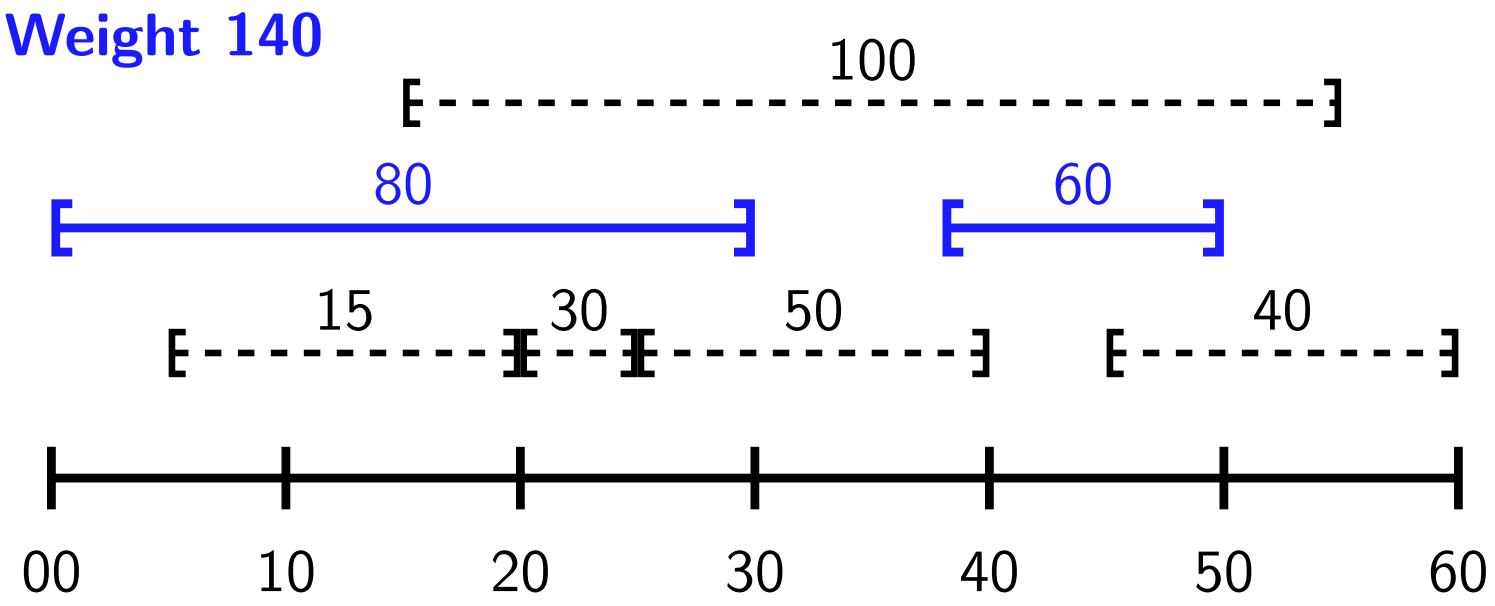

In this case, the desired output has weight 140. But our old greedy algorithm fails! There is **no** known greedy algorithm for this problem.

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Why call it "dynamic programming"?

Because Richard Bellman needed to sell it to an idiot politician and "dynamic" was a fashionable word! If it had been invented today, it would have been called "agile blockchain programming in the cloud"...

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You can think of weighted interval scheduling as a sequence of **choices**: Do I include this interval in my output, or not?

Our greedy algorithm decided "yes" or "no" based only on finishing times. But to reduce a problem to itself, we consider the **effect** of each choice.

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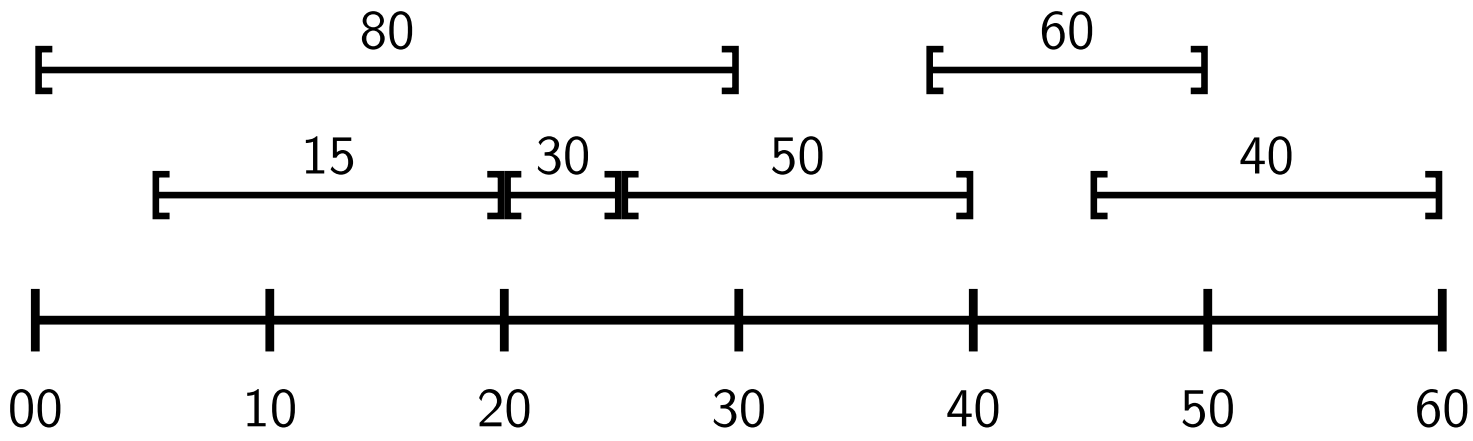

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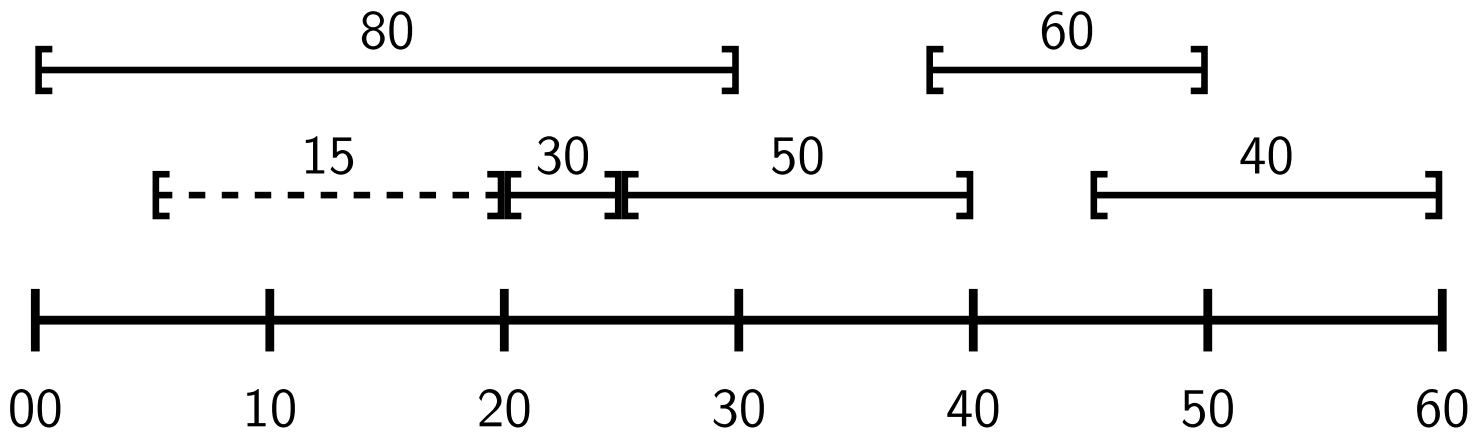

If we don't include some interval *I*: Then the maximum-weight compatible set will be the same as $\mathrm{WIS}(\mathcal{R} \setminus \{I\})$.

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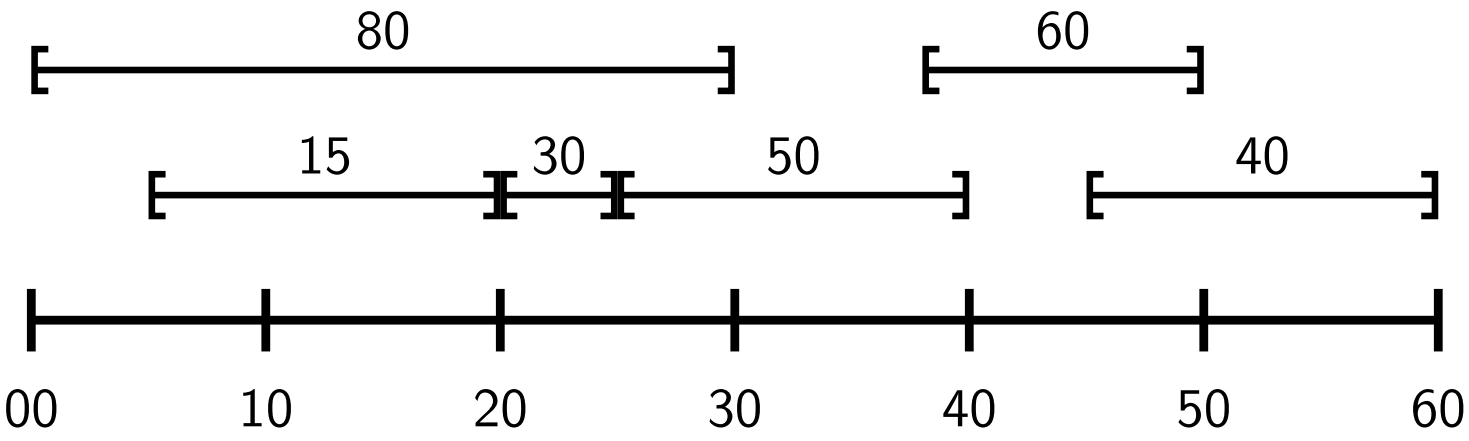

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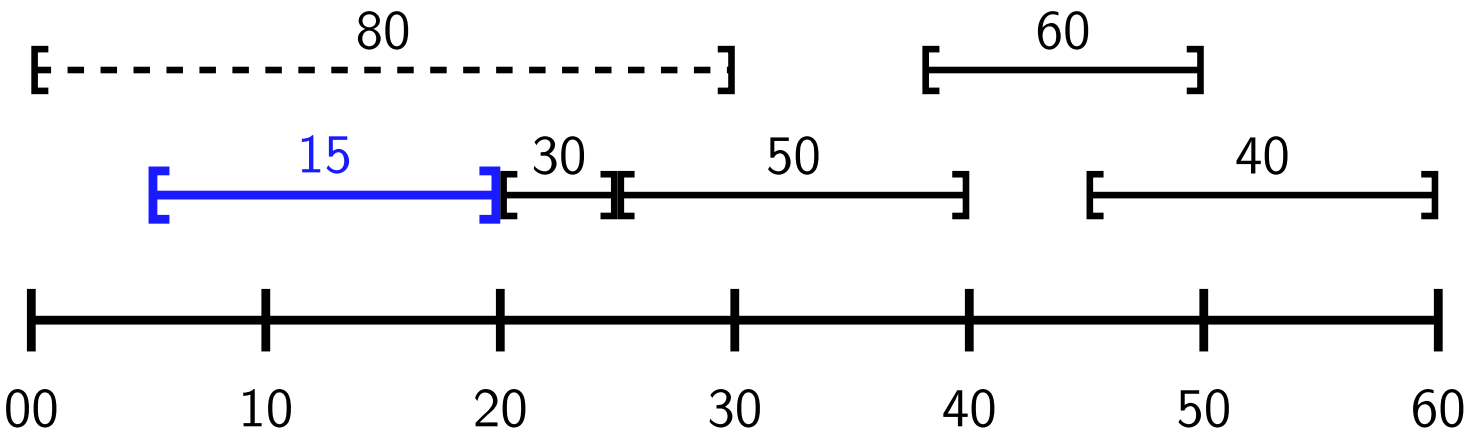

If we do include some interval I: Then we can't include any interval in the set $X_I \subseteq \mathcal{R}$ of intervals intersecting I, or we lose compatibility. But the maximum-weight compatible set will be I together with $\mathrm{WIS}(\mathcal{R} \setminus X_I)$.

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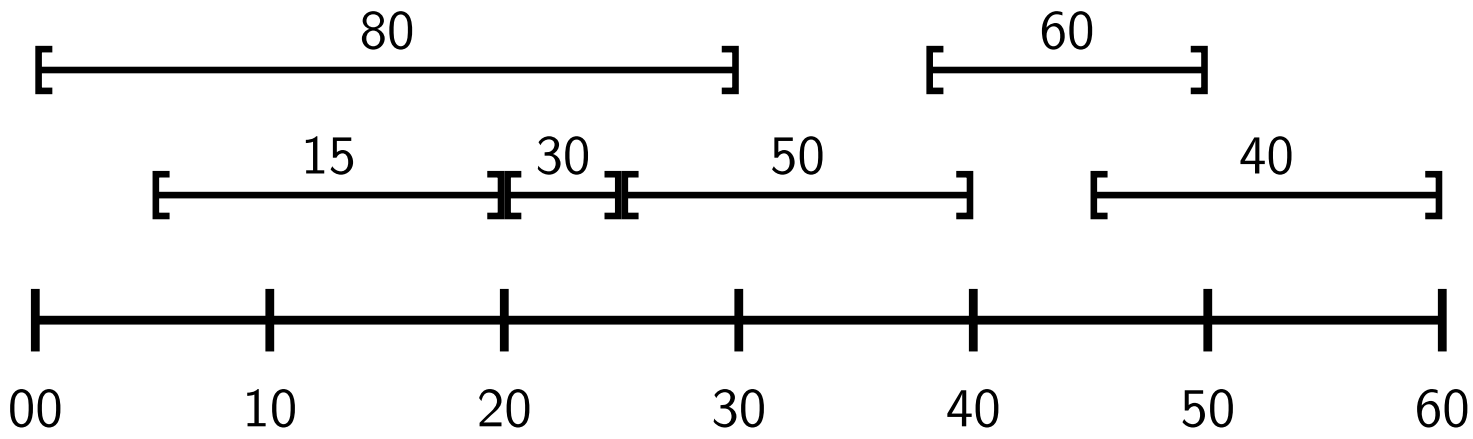

So overall, the highest-weight compatible set will be either $\mathrm{WIS}(\mathcal{R} \setminus \{I\})$ or $\{I\} \cup \mathrm{WIS}(\mathcal{R} \setminus X_I)$, whichever has higher weight.

(See problem sheet 8 question 4 for more examples!)

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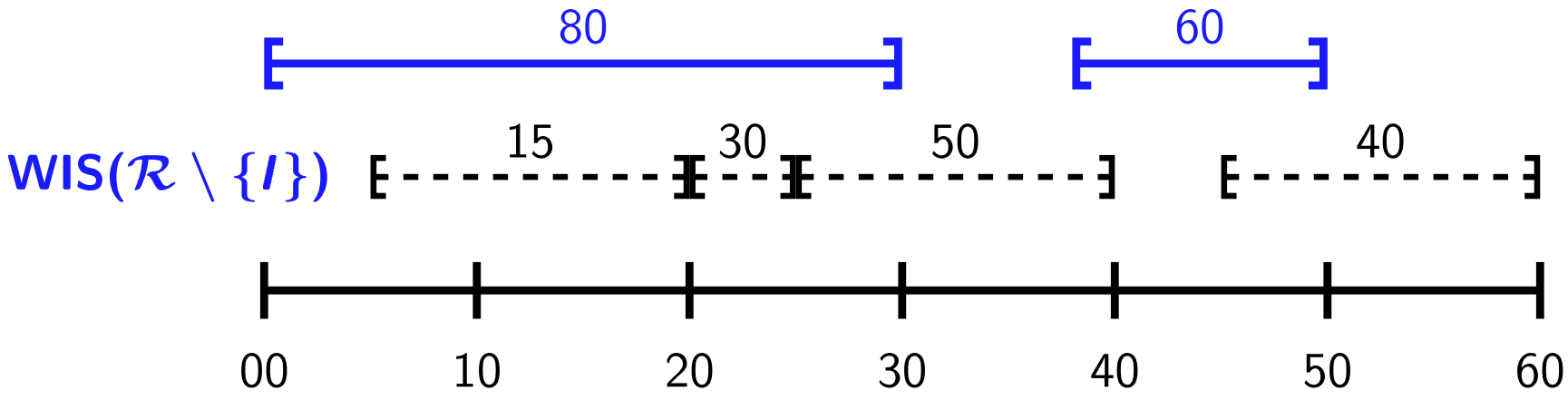

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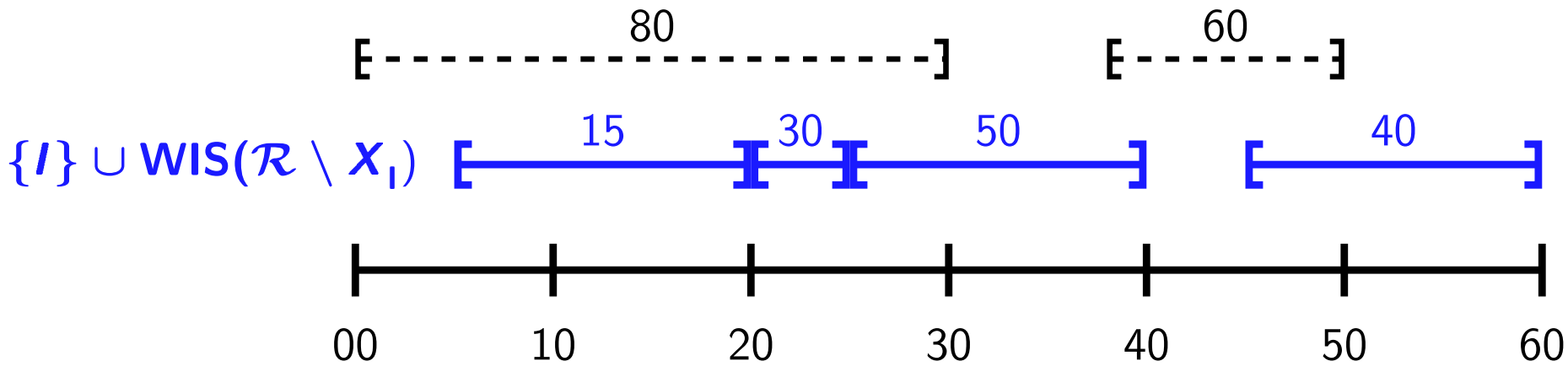

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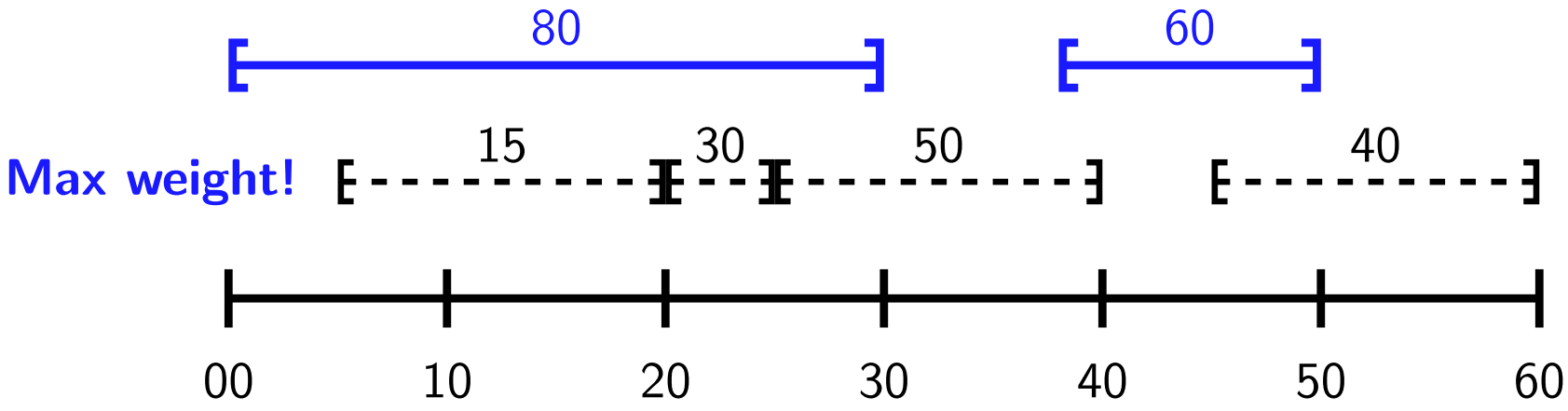

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The full recursive algorithm

Algorithm: WIS

Input : An array $\mathcal{R}$ of n requests and a weight function w.
Output : A maximum-weight compatible subset of $\mathcal{R}$.

```
begin
    if R = ∅ then
        Return ∅.
    else
        Choose I ∈ R arbitrarily.
        Find the set X_I of intervals in R incompatible with I.
        S_out ← WIS(R \ {I}, w).
        S_in ← {I} ∪ WIS(R \ X_I, w).
        if w(S_out) > w(S_in) then
            Return S_out.
        else
            Return S_in.
```

Note that this algorithm will work **regardless** of how we pick each I.

Next video, we will exploit this to make the algorithm run much faster...